MEDICOS MINDS

Better Understanding... Better Care. ♥

NEUROLOGY

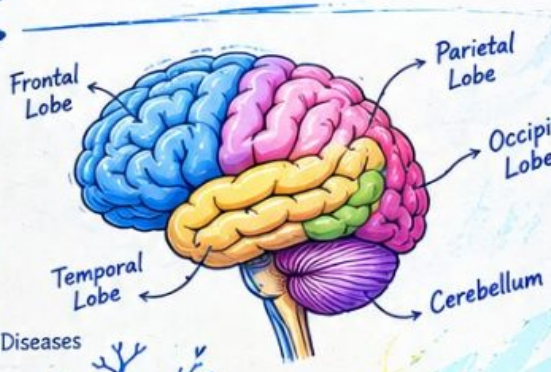
Essentials for Every Medical Student

Understand the Brain
• Diagnose with Confidence • Treat with Care

Same People... Different Stories... Same Goal... HEALTH ♥

KEY TOPICS

- ✓ Neuroanatomy
- ✓ Neuroophysiology
- ✓ CNS & PNS
- ✓ Motor & Sensory Systems
- ✓ Cranial Nerves
- ✓ Neuroimaging (CT / MRI)
- ✓ EEG & EMG
- ✓ Neuro-infectious Diseases
- ✓ Stroke & Cerebrovascular Diseases
- ✓ Movement Disorders
- ✓ Epilepsy
- ✓ Demyelinating Disorders
- ✓ Neurodegenerative Disorders
- ✓ Headache & Facial Pain
- ✓ Tumors
- ✓ Neuro-ICU & Emergency Neurology





Small Circuits... Big Impact! ♥

Think Like a Neurologist ♥

KEY SKILLS

- Neurological Examination
- Interpretation of Imaging
- Clinical Reasoning
- Patient Communication
- Evidence Based Practice

REMEMBER

- ✓ Symptoms are clues...
- ✓ Signs are answers...
- ✓ Neurology is a puzzle...
- ✓ You can solve it! ♥

IMPORTANT CONCEPTS

- Localization of Lesions
- Reflexes & Pathways
- CSF Analysis
- Neuroplasticity
- Autonomic Nervous System
- Approach to Neurological Deficit



ANATOMY
PHYSIOLOGY
PATHOLOGY
CLINICAL
PRACTICE

COMMON CONDITIONS

- ✓ Stroke
- ✓ Epilepsy
- ✓ Parkinson's Disease
- ✓ Alzheimer's Disease
- ✓ Multiple Sclerosis
- ✓ Meningitis / Encephalitis
- ✓ Guillain-Barré Syndrome
- ✓ Neuropathy
- ✓ Brain Tumor



Early Recognition Saves Nerves ♥

ANATOMY
PHYSIOLOGY
PATHOLOGY
CLINICAL
PRACTICE



Better Brain • Better Lives ♥

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✓ Study
✓ Practice
✓ Improve ♥

MEDICOS MINDS

HIGH-YIELD BPT STUDY GUIDE & EXAM COMPANION

Neurology

Author: Dr. MD Kamran Alam | Instagram: @restore_healthphysio

OFFICIAL DIGITAL STUDY GUIDE VERIFICATION

Format: High-Yield Digital PDF | Standard Price: Rs. 59

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STUDY GUIDE DETAILS & CONTENTS:

Comprehensive high-yield notes designed for Bachelor of Physiotherapy (BPT), MBBS, BDS, and allied healthcare students. Covers fundamental anatomy, clinical assessments, pathological conditions, evidence-based physical therapy rehabilitation protocols, and exam-oriented high-yield review.

KEY HIGHLIGHTS:

- Exam-Focused: Concise points curated from leading medical textbooks and clinical references.
- Visual Diagrams: High-yield diagrams and tables for quick revision.
- Mobile & Tablet Optimized: Readable on phones, tablets, iPads, and laptops.
- Official Medicos Minds Release: Verified by Dr. MD Kamran Alam.



NEUROLOGY

INDEX (59 TOPICS)



(Etiology • Pathogenesis • Clinical Features • Investigations • Differential Diagnosis • Management – Medical & Surgical)

No.	Topic
1.	Cerebrovascular disease (Stroke, TIA, RIA)
2.	Types of stroke (Hemorrhagic, Ischemic, Venous infarct)
3.	Stroke syndromes (MCA, PCA, ACA, Vestibulobasilar)
4.	Risk factors, causes, differential diagnosis & management of stroke
5.	Classification of hemorrhagic stroke
6.	Head injury – Etiology, classification, clinical features, investigations, differential diagnosis, management and complications
7.	Spinal cord injury
8.	Spinal epidural abscess
9.	Transverse myelitis
10.	Spina bifida
11.	Sub-acute combined degeneration of cord
12.	Cervical and lumbar disc lesions
13.	Spinal tumors and spinal arachnoiditis
14.	Progressive encephalomyelitis
15.	Psychiatry – Definition, defense mechanism, symptomatology, types & causes of mental disorders
16.	Psychosomatic disorders
17.	Psychosis
18.	Schizophrenia (including paranoid)
19.	Manic depressive psychosis
20.	Demyelinating diseases (central and peripheral)
21.	Guillain – Barre syndrome
22.	Acute disseminated encephalomyelitis
23.	Multiple sclerosis
24.	Parkinson's disease & Kinetic disorders
25.	Dystonia
26.	Myoclonus
27.	Wilson's disease
28.	Infections of brain & spinal cord
29.	Poliomyelitis
30.	Meningitis

No.	Topic
31.	Encephalitis
32.	Brain abscess
33.	Tuberculosis infection of CNS
34.	Complications of systemic infections on nervous system – Rheumatic fever
35.	AIDS
36.	Tetanus
37.	Pertussis
38.	Psychoneurosis – Anxiety
39.	Hysteria
40.	Anxiety states
41.	Neurasthesis
42.	Reactive depression
43.	Obsessive compulsive neurosis
44.	Organic reaction to toxins
45.	Trauma & infection
46.	Senile dementia
47.	Epilepsy – Definition, classification, clinical features, investigations & management
48.	Intracranial tumors – Broad classification, signs & symptoms, investigations & management
49.	Disorders of the muscle
50.	Neuromuscular junction and motor neuron
51.	Muscle diseases – Muscular dystrophy, myotonic dystrophy, myopathy, non-dystrophic myotonia
52.	Disorders of neuromuscular junction – Myasthenia gravis, botulism
53.	Amyotrophic lateral sclerosis
54.	Spinal muscular dystrophy
55.	Neuromyotonia
56.	Mental retardation – Definition, causes, manifestation & management
57.	Methods of treatment in Psychiatry (Brief outline)
58.	Psychotherapy – Group therapy, Psychodrama, Behaviour modification, Family therapy, Play therapy, CBT, REBT, Psychoanalysis, Hypnosis
59.	NLP



Study
Learn
Grow





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SPINAL TUMORS AND SPINAL ARACHNOIDITIS

(For all Medical Students)

MBBS | BHMS | BDS | BSC.NURSING | BPT

Spine matters... Neurology & Orthopaedics both!

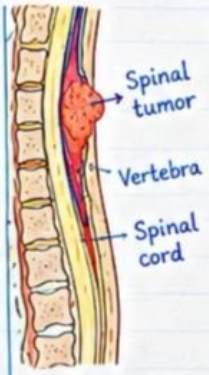
Early Recognition → Better Outcome

SPINAL TUMORS

1. INTRODUCTION

- Tumors can be intramedullary, extramedullary (intradural or extradural) or metastatic.
- May be primary (from spine) or secondary (metastasis).
- Common in adults, but can occur at any age.

Key point: Metastatic tumors are more common than primary tumors in adults.





MRI – shows compressive lesion

2. CLASSIFICATION

A. Primary Spinal Tumors

- **Benign:** Schwannoma, Meningioma, Hemangioma, Osteoma, Osteochondroma, Giant cell tumor.
- **Malignant:** Ependymoma, Astrocytoma, Chordoma, Osteosarcoma, Ewing's sarcoma.

B. Secondary (Metastatic) Tumors

- Common primaries: Breast, Lung, Prostate, Kidney, Thyroid, Melanoma.

3. CLINICAL FEATURES

- Pain (most common, worse at night / on rest).
- Radiculopathy / myelopathy (depending on level).
- Motor weakness, sensory loss, sphincter dysfunction.
- Pathological fracture (in vertebral body lesions).
- Severe cases → paraplegia / quadriplegia.

4. INVESTIGATIONS

- MRI spine – gold standard (shows extent, cord compression).
- CT scan – bony involvement.
- Plain X-ray – lytic / sclerotic lesions, vertebral collapse.
- Biopsy – for histopathological diagnosis.

5. MANAGEMENT

- Surgery – decompression & stabilization (if possible).
- Radiotherapy – for radiosensitive / inoperable tumors.
- Chemotherapy – for malignant / radiosensitive tumors.
- Pain control & rehabilitation.

Early diagnosis + proper treatment can prevent permanent neurological deficit.

SPINAL ARACHNOIDITIS

1. INTRODUCTION

- Inflammation and fibrosis of the arachnoid mater → adhesions, clumping of nerve roots.
- Can be due to infection, trauma, surgery, hemorrhage, or chemical irritation (e.g. contrast, meds).
- Leads to chronic pain and neurological deficits.

2. ETIOLOGY

A. Infective

- TB meningitis
- Bacterial / viral meningitis
- Fungal (rare)

B. Non-infective

- Post-surgical (lumbar)
- Trauma
- Hemorrhage
- Chemical (contrast, drugs)
- Idiopathic (rare)

3. PATHOPHYSIOLOGY

Inflammation of arachnoid → Fibrosis & adhesions of nerve roots → Clumping of cauda equina → Impaired CSF flow & chronic nerve compression

4. CLINICAL FEATURES

- Chronic back pain (worse on movement).
- Radicular pain (burning / electric).
- Neurological deficit (motor, sensory).
- Sphincter dysfunction (late).
- May present with symptoms of raised CSF pressure (headache, nausea).

5. INVESTIGATIONS

- MRI with contrast – nerve root clumping, enhancement, "empty thecal sac" sign.
- CT myelography – shows block / adhesions.
- CSF analysis – ↑ protein, lymphocytic pleocytosis.

6. MANAGEMENT

- Treat underlying cause (anti-TB, antibiotics, etc.).
- Pain management (analgesics, neuropathic pain meds).
- Steroids (in selected cases).
- Physiotherapy – maintain mobility, prevent contractures.
- Surgery – rarely, in severe cases (adhesiolysis).

COMPARISON

Feature	Spinal Tumors	Spinal Arachnoiditis
Nature	Neoplastic growth	Inflammatory / fibrotic
Onset	Slow (months–years)	Variable (subacute–chronic)
Pain	Common (night / rest)	Chronic, radicular
Neurology	Focal deficit	Multiple root involvement
MRI	Mass lesion, enhancement	Clumping, empty thecal sac
CSF	Often normal	↑ Protein, lymphocytes

KEY DIFFERENCE

Tumors form a mass (occupying lesion), while arachnoiditis causes adhesions and nerve root clumping without a discrete mass.

Think Spine... Think Function... Think Early Diagnosis

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THANK YOU FOR YOUR PURCHASE!

We wish you the very best in your medical and physiotherapy examinations.

For any questions, supplementary questions, or study support, reach out to us:

SUPPORT & COMMUNITY:

- Instagram: @restore_healthphysio (Daily Clinical Case Studies & Quizzes)
- WhatsApp Helpline: +91 83407 49923
- Email Support: restorehealthphysio@gmail.com

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